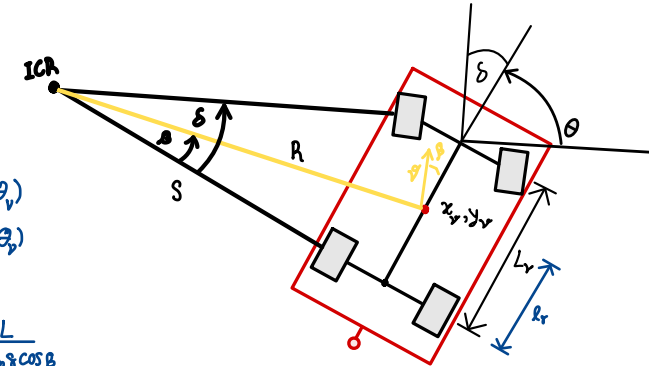




Kinematic Bicycle Model (Desired point is at the center of gravity or CG)



$$\dot{x}_v = v \cos(\beta + \theta_v)$$

$$\dot{y}_v = v \sin(\beta + \theta_v)$$

$$S = \frac{L}{\tan \beta}$$

$$R = \frac{S}{\cos \beta} = \frac{L}{\tan \beta \cos \beta}$$

$$\dot{\theta}_v = \frac{v}{R} = \frac{v}{L} \tan \beta \cos \beta$$

$$\tan \beta = \frac{L_r}{S} = \frac{L_r}{\left(\frac{L}{\tan \beta}\right)} \Rightarrow \beta = \tan^{-1} \left(\frac{L_r \tan \beta}{L} \right)$$

$$x_j = x_v - L_{hitch} \cos \theta_v - L_{bar} \cos \theta_j$$

$$y_j = y_v - L_{hitch} \sin \theta_v - L_{bar} \sin \theta_j$$

$$\dot{x}_j = \frac{dx_j}{dt} = \frac{d}{dt} (x_v - L_{hitch} \cos \theta_v - L_{bar} \cos \theta_j) = \dot{x}_v + L_{hitch} \sin \theta_v \dot{\theta}_v + L_{bar} \sin \theta_j \dot{\theta}_j$$

$$\dot{y}_j = \frac{dy_j}{dt} = \frac{d}{dt} (y_v - L_{hitch} \sin \theta_v - L_{bar} \sin \theta_j) = \dot{y}_v - L_{hitch} \cos \theta_v \dot{\theta}_v - L_{bar} \cos \theta_j \dot{\theta}_j$$

$$x_t = x_v - L_{hitch} \cos \theta_v - L_{bar} \cos \theta_j - L_{trail} \cos \theta_t$$

$$y_t = y_v - L_{hitch} \sin \theta_v - L_{bar} \sin \theta_j - L_{trail} \sin \theta_t$$

$$\begin{aligned} \dot{x}_t &= \frac{d}{dt} (x_v - L_{hitch} \cos \theta_v - L_{bar} \cos \theta_j - L_{trail} \cos \theta_t) \\ &= \dot{x}_v + L_{hitch} \sin \theta_v \dot{\theta}_v + L_{bar} \sin \theta_j \dot{\theta}_j + L_{trail} \sin \theta_t \dot{\theta}_t \end{aligned}$$

$$\begin{aligned} \dot{y}_t &= \frac{d}{dt} (y_v - L_{hitch} \sin \theta_v - L_{bar} \sin \theta_j - L_{trail} \sin \theta_t) \\ &= \dot{y}_v - L_{hitch} \cos \theta_v \dot{\theta}_v - L_{bar} \cos \theta_j \dot{\theta}_j - L_{trail} \cos \theta_t \dot{\theta}_t \end{aligned}$$

Kinetic Energy

The kinetics energy are derive based on Planar Rigid-Body Dynamic. For any rigid body moving on a 2D plane. It total kinetics energy consists of two components

- 1) Transition kinetic Energy: Associated with the absolute linear velocity of the center of mass
- 2) Rotational kinetic Energy: Associated with its rotation (yaw rate) about the vertical z-axis passing through its center of mass

General formula for the kinetic energy T_i of body i is

$$T_i = \frac{1}{2} m_i (\dot{x}_i^2 + \dot{y}_i^2) + \frac{1}{2} I_i \dot{\theta}_i^2$$

Tow Tractor

$$T_t = \frac{1}{2} m_t (\dot{x}_t^2 + \dot{y}_t^2) + \frac{1}{2} I_t \dot{\theta}_t^2$$

Drawbar

$$\begin{aligned} T_d &= \frac{1}{2} m_d (\dot{x}_d^2 + \dot{y}_d^2) + \frac{1}{2} I_d \dot{\theta}_d^2 = \frac{1}{2} m_d \left[(\dot{x}_t + L_{hitcl} \sin \theta_t \dot{\theta}_t + L_{bar} \sin \theta_d \dot{\theta}_d)^2 + (\dot{y}_t - L_{hitcl} \cos \theta_t \dot{\theta}_t - L_{bar} \cos \theta_d \dot{\theta}_d)^2 \right] + \frac{1}{2} I_d \dot{\theta}_d^2 \\ &= \frac{1}{2} m_d \left[(\dot{x}_t^2 + L_{hitcl}^2 \dot{\theta}_t^2 \sin^2 \theta_t + L_{bar}^2 \dot{\theta}_d^2 \sin^2 \theta_d + 2 \dot{x}_t L_{hitcl} \dot{\theta}_t \sin \theta_t + 2 \dot{x}_t L_{bar} \dot{\theta}_d \sin \theta_d + 2 L_{hitcl} L_{bar} \dot{\theta}_t \dot{\theta}_d \sin \theta_t \sin \theta_d) + (\dot{y}_t^2 + L_{hitcl}^2 \dot{\theta}_t^2 \cos^2 \theta_t + L_{bar}^2 \dot{\theta}_d^2 \cos^2 \theta_d - 2 \dot{y}_t L_{hitcl} \dot{\theta}_t \cos \theta_t - 2 \dot{y}_t L_{bar} \dot{\theta}_d \cos \theta_d + 2 L_{hitcl} L_{bar} \dot{\theta}_t \dot{\theta}_d \cos \theta_t \cos \theta_d) \right] + \frac{1}{2} I_d \dot{\theta}_d^2 \\ &= \frac{1}{2} m_d \left[\dot{x}_t^2 + \dot{y}_t^2 + L_{hitcl}^2 \dot{\theta}_t^2 (\sin^2 \theta_t + \cos^2 \theta_t) + L_{bar}^2 \dot{\theta}_d^2 (\sin^2 \theta_d + \cos^2 \theta_d) + 2 L_{hitcl} \dot{\theta}_t (\dot{x}_t \sin \theta_t - \dot{y}_t \cos \theta_t) + 2 L_{bar} \dot{\theta}_d (\dot{x}_t \sin \theta_d - \dot{y}_t \cos \theta_d) + 2 L_{hitcl} L_{bar} \dot{\theta}_t \dot{\theta}_d (\sin \theta_t \sin \theta_d + \cos \theta_t \cos \theta_d) \right] + \frac{1}{2} I_d \dot{\theta}_d^2 \\ &= \frac{1}{2} m_d \left[\dot{x}_t^2 + \dot{y}_t^2 + L_{hitcl}^2 \dot{\theta}_t^2 + L_{bar}^2 \dot{\theta}_d^2 + 2 L_{hitcl} \dot{\theta}_t (\dot{x}_t \sin \theta_t - \dot{y}_t \cos \theta_t) + 2 L_{bar} \dot{\theta}_d (\dot{x}_t \sin \theta_d - \dot{y}_t \cos \theta_d) + 2 L_{hitcl} L_{bar} \dot{\theta}_t \dot{\theta}_d \cos(\theta_t - \theta_d) \right] + \frac{1}{2} I_d \dot{\theta}_d^2 \end{aligned}$$

Trailer

$$\begin{aligned} T_f &= \frac{1}{2} m_f (\dot{x}_f^2 + \dot{y}_f^2) + \frac{1}{2} I_f \dot{\theta}_f^2 = \frac{1}{2} m_f \left[(\dot{x}_t + L_{hitcl} \dot{\theta}_t \sin \theta_t + L_{bar} \dot{\theta}_d \sin \theta_d + L_{trail} \dot{\theta}_f \sin \theta_f) + (\dot{y}_t - L_{hitcl} \cos \theta_t \dot{\theta}_t - L_{bar} \cos \theta_d \dot{\theta}_d - L_{trail} \cos \theta_f \dot{\theta}_f) \right]^2 + \frac{1}{2} I_f \dot{\theta}_f^2 \\ &= \frac{1}{2} m_f \left[(\dot{x}_t^2 + L_{hitcl}^2 \dot{\theta}_t^2 \sin^2 \theta_t + L_{bar}^2 \dot{\theta}_d^2 \sin^2 \theta_d + L_{trail}^2 \dot{\theta}_f^2 \sin^2 \theta_f + 2 L_{hitcl} \dot{\theta}_t \dot{x}_t \sin \theta_t + 2 L_{bar} \dot{\theta}_d \dot{x}_t \sin \theta_d + 2 L_{trail} \dot{\theta}_f \dot{x}_t \sin \theta_f + 2 L_{hitcl} L_{bar} \dot{\theta}_t \dot{\theta}_d \sin \theta_t \sin \theta_d + 2 L_{hitcl} L_{trail} \dot{\theta}_t \dot{\theta}_f \sin \theta_t \sin \theta_f + 2 L_{bar} L_{trail} \dot{\theta}_d \dot{\theta}_f \sin \theta_d \sin \theta_f) \right. \\ &\quad \left. + (\dot{y}_t^2 + L_{hitcl}^2 \dot{\theta}_t^2 \cos^2 \theta_t + L_{bar}^2 \dot{\theta}_d^2 \cos^2 \theta_d + L_{trail}^2 \dot{\theta}_f^2 \cos^2 \theta_f - 2 L_{hitcl} \dot{\theta}_t \dot{y}_t \cos \theta_t - 2 L_{bar} \dot{\theta}_d \dot{y}_t \cos \theta_d - 2 L_{trail} \dot{\theta}_f \dot{y}_t \cos \theta_f + 2 L_{hitcl} L_{bar} \dot{\theta}_t \dot{\theta}_d \cos \theta_t \cos \theta_d + 2 L_{hitcl} L_{trail} \dot{\theta}_t \dot{\theta}_f \cos \theta_t \cos \theta_f + 2 L_{bar} L_{trail} \dot{\theta}_d \dot{\theta}_f \cos \theta_d \cos \theta_f) \right] + \frac{1}{2} I_f \dot{\theta}_f^2 \\ &= \frac{1}{2} m_f \left[(\dot{x}_t^2 + \dot{y}_t^2 + L_{hitcl}^2 \dot{\theta}_t^2 (\sin^2 \theta_t + \cos^2 \theta_t) + L_{bar}^2 \dot{\theta}_d^2 (\sin^2 \theta_d + \cos^2 \theta_d) + L_{trail}^2 \dot{\theta}_f^2 (\sin^2 \theta_f + \cos^2 \theta_f) + 2 L_{hitcl} \dot{\theta}_t (\dot{x}_t \sin \theta_t - \dot{y}_t \cos \theta_t) + 2 L_{bar} \dot{\theta}_d (\dot{x}_t \sin \theta_d - \dot{y}_t \cos \theta_d) + 2 L_{trail} \dot{\theta}_f (\dot{x}_t \sin \theta_f - \dot{y}_t \cos \theta_f) \right. \\ &\quad \left. + 2 L_{hitcl} L_{bar} \dot{\theta}_t \dot{\theta}_d (\sin \theta_t \sin \theta_d + \cos \theta_t \cos \theta_d) + 2 L_{hitcl} L_{trail} \dot{\theta}_t \dot{\theta}_f (\sin \theta_t \sin \theta_f + \cos \theta_t \cos \theta_f) + 2 L_{bar} L_{trail} \dot{\theta}_d \dot{\theta}_f (\sin \theta_d \sin \theta_f + \cos \theta_d \cos \theta_f) \right] + \frac{1}{2} I_f \dot{\theta}_f^2 \\ &= \frac{1}{2} m_f \left[\dot{x}_t^2 + \dot{y}_t^2 + L_{hitcl}^2 \dot{\theta}_t^2 + L_{bar}^2 \dot{\theta}_d^2 + L_{trail}^2 \dot{\theta}_f^2 + 2 L_{hitcl} \dot{\theta}_t (\dot{x}_t \sin \theta_t - \dot{y}_t \cos \theta_t) + 2 L_{bar} \dot{\theta}_d (\dot{x}_t \sin \theta_d - \dot{y}_t \cos \theta_d) + 2 L_{trail} \dot{\theta}_f (\dot{x}_t \sin \theta_f - \dot{y}_t \cos \theta_f) \right. \\ &\quad \left. + 2 L_{hitcl} L_{bar} \dot{\theta}_t \dot{\theta}_d \cos(\theta_t - \theta_d) + 2 L_{hitcl} L_{trail} \dot{\theta}_t \dot{\theta}_f \cos(\theta_t - \theta_f) + 2 L_{bar} L_{trail} \dot{\theta}_d \dot{\theta}_f \cos(\theta_d - \theta_f) \right] + \frac{1}{2} I_f \dot{\theta}_f^2 \end{aligned}$$

Total kinetic Energy

$$T = T_v + T_\theta + T_\phi$$

$$= \left[\frac{1}{2} m_v (\dot{x}_v^2 + \dot{y}_v^2) + \frac{1}{2} I_v \dot{\theta}_v^2 \right] + \left[\frac{1}{2} m_\theta \left[\dot{x}_\theta^2 + \dot{y}_\theta^2 + L_{hitch}^2 \dot{\theta}_\theta^2 + L_{bar}^2 \dot{\theta}_\theta^2 + 2 L_{hitch} \dot{\theta}_\theta (\dot{x}_\theta \sin \theta_\theta - \dot{y}_\theta \cos \theta_\theta) + 2 L_{bar} \dot{\theta}_\theta (\dot{x}_\theta \sin \theta_\theta - \dot{y}_\theta \cos \theta_\theta) + 2 L_{hitch} L_{bar} \dot{\theta}_\theta \dot{\theta}_\theta \cos(\theta_\theta - \theta_\phi) \right] + \frac{1}{2} I_\theta \dot{\theta}_\theta^2 \right] + \left[\frac{1}{2} m_\phi \left[\dot{x}_\phi^2 + \dot{y}_\phi^2 + L_{hitch}^2 \dot{\theta}_\phi^2 + L_{bar}^2 \dot{\theta}_\phi^2 + L_{trail}^2 \dot{\theta}_\phi^2 + 2 L_{hitch} \dot{\theta}_\phi (\dot{x}_\phi \sin \theta_\phi - \dot{y}_\phi \cos \theta_\phi) + 2 L_{bar} \dot{\theta}_\phi (\dot{x}_\phi \sin \theta_\phi - \dot{y}_\phi \cos \theta_\phi) + 2 L_{trail} \dot{\theta}_\phi (\dot{x}_\phi \sin \theta_\phi - \dot{y}_\phi \cos \theta_\phi) + 2 L_{hitch} L_{bar} \dot{\theta}_\phi \dot{\theta}_\phi \cos(\theta_\phi - \theta_\theta) + 2 L_{hitch} L_{trail} \dot{\theta}_\phi \dot{\theta}_\phi \cos(\theta_\phi - \theta_\theta) + 2 L_{bar} L_{trail} \dot{\theta}_\phi \dot{\theta}_\phi \cos(\theta_\phi - \theta_\theta) \right] + \frac{1}{2} I_\phi \dot{\theta}_\phi^2 \right]$$

$$= \frac{1}{2} (m_v + m_\theta + m_\phi) \dot{x}_v^2 + \frac{1}{2} (m_v + m_\theta + m_\phi) \dot{y}_v^2 + \frac{1}{2} [I_v + (m_\theta + m_\phi) L_{hitch}^2] \dot{\theta}_v^2 + \frac{1}{2} [I_v + (m_\theta + m_\phi) L_{bar}^2] \dot{\theta}_\theta^2 + \frac{1}{2} [I_\phi + (m_\theta + m_\phi) L_{trail}^2] \dot{\theta}_\phi^2 + (m_\theta + m_\phi) L_{hitch} \dot{\theta}_v (\dot{x}_v \sin \theta_v - \dot{y}_v \cos \theta_v) + (m_\theta + m_\phi) L_{bar} \dot{\theta}_v (\dot{x}_v \sin \theta_v - \dot{y}_v \cos \theta_v) + (m_\theta + m_\phi) L_{hitch} L_{bar} \dot{\theta}_v \dot{\theta}_\theta \cos(\theta_v - \theta_\theta) + (m_\theta + m_\phi) L_{hitch} L_{trail} \dot{\theta}_v \dot{\theta}_\phi \cos(\theta_v - \theta_\phi) + (m_\theta + m_\phi) L_{bar} L_{trail} \dot{\theta}_v \dot{\theta}_\phi \cos(\theta_\theta - \theta_\phi)$$

Total Potential Energy

$$V = 0$$

Euler Lagrangian

$$L = T + V$$

$$L = T$$

$$L = \frac{1}{2} (m_v + m_\theta + m_\phi) \dot{x}_v^2 + \frac{1}{2} (m_v + m_\theta + m_\phi) \dot{y}_v^2 + \frac{1}{2} [I_v + (m_\theta + m_\phi) L_{hitch}^2] \dot{\theta}_v^2 + \frac{1}{2} [I_v + (m_\theta + m_\phi) L_{bar}^2] \dot{\theta}_\theta^2 + \frac{1}{2} [I_\phi + (m_\theta + m_\phi) L_{trail}^2] \dot{\theta}_\phi^2 + (m_\theta + m_\phi) L_{hitch} \dot{\theta}_v (\dot{x}_v \sin \theta_v - \dot{y}_v \cos \theta_v) + (m_\theta + m_\phi) L_{bar} \dot{\theta}_v (\dot{x}_v \sin \theta_v - \dot{y}_v \cos \theta_v) + (m_\theta + m_\phi) L_{hitch} L_{bar} \dot{\theta}_v \dot{\theta}_\theta \cos(\theta_v - \theta_\theta) + (m_\theta + m_\phi) L_{hitch} L_{trail} \dot{\theta}_v \dot{\theta}_\phi \cos(\theta_v - \theta_\phi) + (m_\theta + m_\phi) L_{bar} L_{trail} \dot{\theta}_v \dot{\theta}_\phi \cos(\theta_\theta - \theta_\phi)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i \quad \text{Generalized Coordinate ; } q = \begin{bmatrix} x_v \\ y_v \\ \theta_v \\ \theta_\theta \\ \theta_\phi \end{bmatrix} ; \quad \dot{q} = \begin{bmatrix} \dot{x}_v \\ \dot{y}_v \\ \dot{\theta}_v \\ \dot{\theta}_\theta \\ \dot{\theta}_\phi \end{bmatrix}$$

$$\text{for } q_i = x_v ; \quad \dot{q}_i = \dot{x}_v$$

$$\frac{\partial L}{\partial q_i} = \frac{\partial L}{\partial x_v} = 0$$

$$\frac{\partial L}{\partial \dot{x}_v} = \frac{\partial}{\partial \dot{x}_v} \left[\frac{1}{2} (m_v + m_\theta + m_\phi) \dot{x}_v^2 \right] + \frac{\partial}{\partial \dot{x}_v} \left[\frac{1}{2} (m_v + m_\theta + m_\phi) \dot{y}_v^2 \right] + \frac{\partial}{\partial \dot{x}_v} \left[\frac{1}{2} [I_v + (m_\theta + m_\phi) L_{hitch}^2] \dot{\theta}_v^2 \right] + \frac{\partial}{\partial \dot{x}_v} \left[\frac{1}{2} [I_v + (m_\theta + m_\phi) L_{bar}^2] \dot{\theta}_\theta^2 \right] + \frac{\partial}{\partial \dot{x}_v} \left[\frac{1}{2} [I_\phi + (m_\theta + m_\phi) L_{trail}^2] \dot{\theta}_\phi^2 \right] + \frac{\partial}{\partial \dot{x}_v} \left[(m_\theta + m_\phi) L_{hitch} \dot{\theta}_v (\dot{x}_v \sin \theta_v - \dot{y}_v \cos \theta_v) \right] + \frac{\partial}{\partial \dot{x}_v} \left[(m_\theta + m_\phi) L_{bar} \dot{\theta}_v (\dot{x}_v \sin \theta_v - \dot{y}_v \cos \theta_v) \right] + \frac{\partial}{\partial \dot{x}_v} \left[(m_\theta + m_\phi) L_{hitch} L_{bar} \dot{\theta}_v \dot{\theta}_\theta \cos(\theta_v - \theta_\theta) \right] + \frac{\partial}{\partial \dot{x}_v} \left[(m_\theta + m_\phi) L_{hitch} L_{trail} \dot{\theta}_v \dot{\theta}_\phi \cos(\theta_v - \theta_\phi) \right] + \frac{\partial}{\partial \dot{x}_v} \left[(m_\theta + m_\phi) L_{bar} L_{trail} \dot{\theta}_v \dot{\theta}_\phi \cos(\theta_\theta - \theta_\phi) \right]$$

$$= (m_v + m_\theta + m_\phi) \dot{x}_v + (m_\theta + m_\phi) L_{hitch} \dot{\theta}_v \sin \theta_v + (m_\theta + m_\phi) L_{bar} \dot{\theta}_v \sin \theta_v + m_\theta L_{hitch} \dot{\theta}_\theta \sin \theta_\theta + m_\phi L_{hitch} \dot{\theta}_\phi \sin \theta_\phi$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_v} \right) - \frac{\partial L}{\partial x_v} = (m_v + m_\theta + m_\phi) \ddot{x}_v + (m_\theta + m_\phi) L_{hitch} (\ddot{\theta}_v \sin \theta_v + \dot{\theta}_v^2 \cos \theta_v) + (m_\theta + m_\phi) L_{bar} (\ddot{\theta}_v \sin \theta_v + \dot{\theta}_v^2 \cos \theta_v) + m_\theta L_{hitch} (\ddot{\theta}_\theta \sin \theta_\theta + \dot{\theta}_\theta^2 \cos \theta_\theta) + m_\phi L_{hitch} (\ddot{\theta}_\phi \sin \theta_\phi + \dot{\theta}_\phi^2 \cos \theta_\phi) = Q_x$$

for $q_i = y_r \rightarrow \dot{q}_i = \dot{y}_r$

$$\frac{\partial L}{\partial q} = \frac{\partial L}{\partial y_r} = 0$$

$$\frac{\partial L}{\partial \dot{y}_P} = \frac{\partial}{\partial \dot{y}_P} \left[\frac{1}{2} (m_P + m_2 + m_4) \dot{x}_P^2 \right] + \frac{\partial}{\partial \dot{y}_P} \left[\frac{1}{2} (m_P + m_2 + m_4) \dot{y}_P^2 \right] + \frac{\partial}{\partial \dot{y}_P} \left[\frac{1}{2} [I_P + (m_2 + m_4) L_{hitd}^2] \dot{\theta}_P^2 \right] + \frac{\partial}{\partial \dot{y}_P} \left[\frac{1}{2} [I_P + (m_2 + m_4) L_{bar}^2] \dot{\theta}_2^2 \right] + \frac{\partial}{\partial \dot{y}_P} \left[\frac{1}{2} [I_P + m_4 L_{trail}^2] \dot{\theta}_4^2 \right] + \frac{\partial}{\partial \dot{y}_P} \left[(m_2 + m_4) L_{hitd} \dot{\theta}_P (\dot{x}_P \sin \theta_P - \dot{y}_P \cos \theta_P) \right] \\ + \frac{\partial}{\partial \dot{y}_P} \left[(m_2 + m_4) L_{bar} \dot{\theta}_2 (\dot{x}_P \sin \theta_2 - \dot{y}_P \cos \theta_2) \right] + \frac{\partial}{\partial \dot{y}_P} \left[m_4 L_{trail} \dot{\theta}_4 (\dot{x}_P \sin \theta_4 - \dot{y}_P \cos \theta_4) \right] + \frac{\partial}{\partial \dot{y}_P} \left[(m_2 + m_4) L_{hitd} L_{bar} \dot{\theta}_P \dot{\theta}_2 \cos(\theta_P - \theta_2) \right] + \frac{\partial}{\partial \dot{y}_P} \left[m_4 L_{hitd} L_{trail} \dot{\theta}_P \dot{\theta}_4 \cos(\theta_P - \theta_4) \right] + \frac{\partial}{\partial \dot{y}_P} \left[m_4 L_{bar} L_{trail} \dot{\theta}_2 \dot{\theta}_4 \cos(\theta_2 - \theta_4) \right] \\ \frac{\partial L}{\partial y_P} = (m_P + m_2 + m_4) \dot{y}_P - (m_2 + m_4) L_{hitd} \dot{\theta}_P \cos \theta_P - (m_2 + m_4) L_{bar} \dot{\theta}_2 \cos \theta_2 - m_4 L_{trail} \dot{\theta}_4 \cos \theta_4$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}_y} \right) - \frac{\partial L}{\partial y_y} = (m_{\text{pr}} + m_j + m_t) \ddot{y}_y - (m_j + m_t) L_{\text{hitch}} (\ddot{\theta}_y \cos \theta_y - \dot{\theta}_y^2 \sin \theta_y) - (m_j + m_t) L_{\text{bay}} (\ddot{\theta}_j \cos \theta_j - \dot{\theta}_j^2 \sin \theta_j) - m_t L_{\text{truss}} (\ddot{\theta}_t \cos \theta_t - \dot{\theta}_t^2 \sin \theta_t) = Q_y$$

for $q_i = \theta_r \rightarrow \dot{q}_i = \dot{\theta}_r$

$$\begin{aligned} \frac{\partial L}{\partial \dot{\theta}_p} &= \frac{\partial L}{\partial \dot{\theta}_p} = \frac{\partial}{\partial \dot{\theta}_p} \left[\frac{1}{2} (m_p + m_s + m_t) \dot{x}_p^2 \right] + \frac{\partial}{\partial \dot{\theta}_p} \left[\frac{1}{2} (m_p + m_s + m_t) \dot{y}_p^2 \right] + \frac{\partial}{\partial \dot{\theta}_p} \left[\frac{1}{2} [I_p + (m_s + m_t) L_{hit}^2] \dot{\theta}_p^2 \right] + \frac{\partial}{\partial \dot{\theta}_p} \left[\frac{1}{2} [I_p + (m_s + m_t) L_{bar}^2] \dot{\theta}_s^2 \right] + \frac{\partial}{\partial \dot{\theta}_p} \left[\frac{1}{2} [I_t + m_t L_{trail}^2] \dot{\theta}_t^2 \right] + \frac{\partial}{\partial \dot{\theta}_p} \left[(m_s + m_t) L_{hit} \dot{\theta}_p (\dot{x}_p \sin \theta_p - \dot{y}_p \cos \theta_p) \right] \\ &\quad + \frac{\partial}{\partial \dot{\theta}_p} \left[(m_s + m_t) L_{bar} \dot{\theta}_s (\dot{x}_p \sin \theta_s - \dot{y}_p \cos \theta_s) \right] + \frac{\partial}{\partial \dot{\theta}_p} \left[m_t L_{trail} \dot{\theta}_t (\dot{x}_p \sin \theta_t - \dot{y}_p \cos \theta_t) \right] + \frac{\partial}{\partial \dot{\theta}_p} \left[(m_s + m_t) L_{hit} L_{bar} \dot{\theta}_p \dot{\theta}_s (\cos(\theta_p - \theta_s)) \right] + \frac{\partial}{\partial \dot{\theta}_p} \left[m_t L_{hit} L_{trail} \dot{\theta}_p \dot{\theta}_t \cos(\theta_p - \theta_t) \right] + \frac{\partial}{\partial \dot{\theta}_p} \left[m_t L_{bar} L_{trail} \dot{\theta}_s \dot{\theta}_t \cos(\theta_s - \theta_t) \right] \\ &= (m_s + m_t) L_{hit} \dot{\theta}_p (\dot{x}_p \cos \theta_p + \dot{y}_p \sin \theta_p) - (m_s + m_t) L_{hit} L_{bar} \dot{\theta}_p \dot{\theta}_s \sin(\theta_p - \theta_s) - m_t L_{hit} L_{trail} \dot{\theta}_p \dot{\theta}_t \sin(\theta_p - \theta_t) \\ \frac{\partial L}{\partial \dot{\theta}_s} &= \frac{\partial}{\partial \dot{\theta}_s} \left[\frac{1}{2} (m_p + m_s + m_t) \dot{x}_p^2 \right] + \frac{\partial}{\partial \dot{\theta}_s} \left[\frac{1}{2} (m_p + m_s + m_t) \dot{y}_p^2 \right] + \frac{\partial}{\partial \dot{\theta}_s} \left[\frac{1}{2} [I_p + (m_s + m_t) L_{hit}^2] \dot{\theta}_p^2 \right] + \frac{\partial}{\partial \dot{\theta}_s} \left[\frac{1}{2} [I_p + (m_s + m_t) L_{bar}^2] \dot{\theta}_s^2 \right] + \frac{\partial}{\partial \dot{\theta}_s} \left[\frac{1}{2} [I_t + m_t L_{trail}^2] \dot{\theta}_t^2 \right] + \frac{\partial}{\partial \dot{\theta}_s} \left[(m_s + m_t) L_{hit} \dot{\theta}_p (\dot{x}_p \sin \theta_p - \dot{y}_p \cos \theta_p) \right] \\ &\quad + \frac{\partial}{\partial \dot{\theta}_s} \left[(m_s + m_t) L_{bar} \dot{\theta}_s (\dot{x}_p \sin \theta_s - \dot{y}_p \cos \theta_s) \right] + \frac{\partial}{\partial \dot{\theta}_s} \left[m_t L_{trail} \dot{\theta}_t (\dot{x}_p \sin \theta_t - \dot{y}_p \cos \theta_t) \right] + \frac{\partial}{\partial \dot{\theta}_s} \left[(m_s + m_t) L_{hit} L_{bar} \dot{\theta}_p \dot{\theta}_s (\cos(\theta_p - \theta_s)) \right] + \frac{\partial}{\partial \dot{\theta}_s} \left[m_t L_{hit} L_{trail} \dot{\theta}_p \dot{\theta}_t \cos(\theta_p - \theta_t) \right] + \frac{\partial}{\partial \dot{\theta}_s} \left[m_t L_{bar} L_{trail} \dot{\theta}_s \dot{\theta}_t \cos(\theta_s - \theta_t) \right] \\ &= [I_p + (m_s + m_t) L_{hit}^2] \dot{\theta}_p + (m_s + m_t) L_{hit} (\dot{x}_p \sin \theta_p - \dot{y}_p \cos \theta_p) + (m_s + m_t) L_{hit} L_{bar} \dot{\theta}_p \dot{\theta}_s (\cos(\theta_p - \theta_s)) + m_t L_{hit} L_{trail} \dot{\theta}_p \dot{\theta}_t \cos(\theta_p - \theta_t) \\ \frac{\partial L}{\partial \dot{\theta}_t} &= [I_p + (m_s + m_t) L_{hit}^2] \dot{\theta}_p + (m_s + m_t) L_{hit} (\dot{x}_p \sin \theta_p - \dot{y}_p \cos \theta_p) + \dot{x}_p \cos \theta_p \dot{\theta}_p - \dot{y}_p \cos \theta_p + \dot{y}_p \sin \theta_p \dot{\theta}_p + (m_s + m_t) L_{hit} L_{bar} [\dot{\theta}_s \cos(\theta_p - \theta_s) - \dot{\theta}_s \dot{\theta}_p \sin(\theta_p - \theta_s) + \dot{\theta}_s^2 \sin(\theta_p - \theta_s)] \\ &\quad + m_t L_{hit} L_{trail} [\dot{\theta}_t \cos(\theta_p - \theta_t) - \dot{\theta}_t \dot{\theta}_p + \dot{\theta}_t^2 \sin(\theta_p - \theta_t)] \end{aligned}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_p} \right) - \frac{\partial L}{\partial \theta_p} = \left[L_p + (m_2 + m_4) L_{hit}^2 \right] \ddot{\theta}_p + (m_2 + m_4) L_{hit} (\dot{x}_p \sin \theta_p - \dot{y}_p \cos \theta_p + \dot{\theta}_p (\dot{x}_p \cos \theta_p + \dot{y}_p \sin \theta_p)) + (m_2 + m_4) L_{hit} L_{bar} [\ddot{\theta}_2 \cos(\theta_p - \theta_2) - \dot{\theta}_2 \dot{\theta}_p \sin(\theta_p - \theta_2) + \dot{\theta}_2^2 \sin(\theta_p - \theta_2)] \\ + m_4 L_{hit} L_{bar} [\ddot{\theta}_4 \cos(\theta_p - \theta_4) - \dot{\theta}_4 \dot{\theta}_p \sin(\theta_p - \theta_4) + \dot{\theta}_4^2 \sin(\theta_p - \theta_4)] - (m_2 + m_4) L_{hit} \ddot{\theta}_p (\dot{x}_p \cos \theta_p + \dot{y}_p \sin \theta_p) + (m_2 + m_4) L_{hit} L_{bar} \dot{\theta}_p \dot{\theta}_2 \sin(\theta_p - \theta_2) + m_4 L_{hit} L_{bar} \dot{\theta}_p \dot{\theta}_4 \sin(\theta_p - \theta_4)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_p} \right) - \frac{\partial L}{\partial \theta_p} = \left[L_p + (m_d + m_f) L_{hi}^2 \right] \ddot{\theta}_p + (m_d + m_f) L_{hi} \dot{\theta}_p (\ddot{x}_p \sin \theta_p - \dot{x}_p \cos \theta_p) + (m_d + m_f) L_{hi} \dot{\theta}_p \ddot{\theta}_d \cos(\theta_p - \theta_d) + (m_d + m_f) L_{hi} \dot{\theta}_p \dot{\theta}_d^2 \sin(\theta_p - \theta_d) + m_f L_{hi} \dot{\theta}_p \ddot{\theta}_f \cos(\theta_p - \theta_f) + m_f L_{hi} \dot{\theta}_p \dot{\theta}_f^2 \sin(\theta_p - \theta_f) = Q_p$$

$$\text{for } q_i = \theta_i \rightarrow \dot{q}_i = \dot{\theta}_i$$

$$\begin{aligned} \frac{\partial L}{\partial \dot{\theta}_i} = \frac{\partial L}{\partial \dot{\theta}_i} &= \frac{\partial}{\partial \dot{\theta}_i} \left(\frac{1}{2} (m_w + m_s + m_t) \dot{x}_w^2 \right) + \frac{\partial}{\partial \dot{\theta}_i} \left(\frac{1}{2} (m_w + m_s + m_t) \dot{y}_w^2 \right) + \frac{\partial}{\partial \dot{\theta}_i} \left(\frac{1}{2} [I_w + (m_s + m_t) L_{hitch}^2] \dot{\theta}_w^2 \right) + \frac{\partial}{\partial \dot{\theta}_i} \left(\frac{1}{2} [I_d + (m_s + m_t) L_{bar}^2] \dot{\theta}_s^2 \right) + \frac{\partial}{\partial \dot{\theta}_i} \left(\frac{1}{2} [I_t + m_t L_{trail}^2] \dot{\theta}_t^2 \right) + \frac{\partial}{\partial \dot{\theta}_i} \left(c(m_s + m_t) L_{hitch} \dot{\theta}_w (\dot{x}_w \sin \theta_w - \dot{y}_w \cos \theta_w) \right) \\ &+ \frac{\partial}{\partial \dot{\theta}_i} \left(c(m_s + m_t) L_{bar} \dot{\theta}_s (\dot{x}_w \sin \theta_s - \dot{y}_w \cos \theta_s) \right) + \frac{\partial}{\partial \dot{\theta}_i} \left(m_t L_{trail} \dot{\theta}_t (\dot{x}_w \sin \theta_t - \dot{y}_w \cos \theta_t) \right) + \frac{\partial}{\partial \dot{\theta}_i} \left((m_s + m_t) L_{hitch} L_{bar} \dot{\theta}_w \dot{\theta}_s (\cos(\theta_w - \theta_s)) \right) + \frac{\partial}{\partial \dot{\theta}_i} \left(m_t L_{hitch} L_{trail} \dot{\theta}_w \dot{\theta}_t (\cos(\theta_w - \theta_t)) \right) + \frac{\partial}{\partial \dot{\theta}_i} \left(m_t L_{bar} L_{trail} \dot{\theta}_s \dot{\theta}_t (\cos(\theta_s - \theta_t)) \right) \\ &= m_t L_{trail} \dot{\theta}_t (\dot{x}_w \cos \theta_s + \dot{y}_w \sin \theta_t) + m_t L_{hitch} L_{trail} \dot{\theta}_w \dot{\theta}_t \sin(\theta_w - \theta_t) + m_t L_{bar} L_{trail} \dot{\theta}_s \dot{\theta}_t \sin(\theta_s - \theta_t) \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial \dot{\theta}_w} &= \frac{\partial}{\partial \dot{\theta}_w} \left(\frac{1}{2} (m_w + m_s + m_t) \dot{x}_w^2 \right) + \frac{\partial}{\partial \dot{\theta}_w} \left(\frac{1}{2} (m_w + m_s + m_t) \dot{y}_w^2 \right) + \frac{\partial}{\partial \dot{\theta}_w} \left(\frac{1}{2} [I_w + (m_s + m_t) L_{hitch}^2] \dot{\theta}_w^2 \right) + \frac{\partial}{\partial \dot{\theta}_w} \left(\frac{1}{2} [I_d + (m_s + m_t) L_{bar}^2] \dot{\theta}_s^2 \right) + \frac{\partial}{\partial \dot{\theta}_w} \left(\frac{1}{2} [I_t + m_t L_{trail}^2] \dot{\theta}_t^2 \right) + \frac{\partial}{\partial \dot{\theta}_w} \left(c(m_s + m_t) L_{hitch} \dot{\theta}_w (\dot{x}_w \sin \theta_w - \dot{y}_w \cos \theta_w) \right) \\ &+ \frac{\partial}{\partial \dot{\theta}_w} \left(c(m_s + m_t) L_{bar} \dot{\theta}_s (\dot{x}_w \sin \theta_s - \dot{y}_w \cos \theta_s) \right) + \frac{\partial}{\partial \dot{\theta}_w} \left(m_t L_{trail} \dot{\theta}_t (\dot{x}_w \sin \theta_t - \dot{y}_w \cos \theta_t) \right) + \frac{\partial}{\partial \dot{\theta}_w} \left((m_s + m_t) L_{hitch} L_{bar} \dot{\theta}_w \dot{\theta}_s (\cos(\theta_w - \theta_s)) \right) + \frac{\partial}{\partial \dot{\theta}_w} \left(m_t L_{hitch} L_{trail} \dot{\theta}_w \dot{\theta}_t (\cos(\theta_w - \theta_t)) \right) + \frac{\partial}{\partial \dot{\theta}_w} \left(m_t L_{bar} L_{trail} \dot{\theta}_s \dot{\theta}_t (\cos(\theta_s - \theta_t)) \right) \\ &= [I_t + m_t L_{trail}^2] \dot{\theta}_t + m_t L_{trail} (\dot{x}_w \sin \theta_t - \dot{y}_w \cos \theta_t) + m_t L_{hitch} L_{trail} \dot{\theta}_w \cos(\theta_w - \theta_t) + m_t L_{bar} L_{trail} \dot{\theta}_s (\cos(\theta_s - \theta_t)) \end{aligned}$$

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{\theta}_w} \right] = [I_t + m_t L_{trail}^2] \ddot{\theta}_t + m_t L_{trail} [\ddot{x}_w \sin \theta_t - \ddot{y}_w \cos \theta_t + \dot{\theta}_t (\dot{x}_w \cos \theta_t + \dot{y}_w \sin \theta_t)] + m_t L_{hitch} L_{trail} (\ddot{\theta}_w \cos(\theta_w - \theta_t) - \dot{\theta}_w^2 \sin(\theta_w - \theta_t) + \dot{\theta}_w \dot{\theta}_t \sin(\theta_w - \theta_t)) + m_t L_{bar} L_{trail} (\ddot{\theta}_s \cos(\theta_s - \theta_t) - \dot{\theta}_s^2 \sin(\theta_s - \theta_t) + \dot{\theta}_s \dot{\theta}_t \sin(\theta_s - \theta_t))$$

$$\begin{aligned} \frac{d}{dt} \left[\frac{\partial L}{\partial \dot{\theta}_w} \right] - \frac{\partial L}{\partial \theta_i} &= [I_t + m_t L_{trail}^2] \ddot{\theta}_t + m_t L_{trail} [\ddot{x}_w \sin \theta_t - \ddot{y}_w \cos \theta_t + \dot{\theta}_t (\dot{x}_w \cos \theta_t + \dot{y}_w \sin \theta_t)] + m_t L_{hitch} L_{trail} (\ddot{\theta}_w \cos(\theta_w - \theta_t) - \dot{\theta}_w^2 \sin(\theta_w - \theta_t) + \dot{\theta}_w \dot{\theta}_t \sin(\theta_w - \theta_t)) + m_t L_{bar} L_{trail} (\ddot{\theta}_s \cos(\theta_s - \theta_t) - \dot{\theta}_s^2 \sin(\theta_s - \theta_t) + \dot{\theta}_s \dot{\theta}_t \sin(\theta_s - \theta_t)) \\ &- m_t L_{trail} \dot{\theta}_t (\dot{x}_w \cos \theta_s + \dot{y}_w \sin \theta_t) - m_t L_{hitch} L_{trail} \dot{\theta}_w \dot{\theta}_t \sin(\theta_w - \theta_t) - m_t L_{bar} L_{trail} \dot{\theta}_s \dot{\theta}_t \sin(\theta_s - \theta_t) \end{aligned}$$

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{\theta}_w} \right] - \frac{\partial L}{\partial \theta_i} = [I_t + m_t L_{trail}^2] \ddot{\theta}_t + m_t L_{trail} (\ddot{x}_w \sin \theta_t - \ddot{y}_w \cos \theta_t) + m_t L_{hitch} L_{trail} \ddot{\theta}_w \cos(\theta_w - \theta_t) + m_t L_{bar} L_{trail} \ddot{\theta}_s \cos(\theta_s - \theta_t) - m_t L_{hitch} L_{trail} \dot{\theta}_w^2 \sin(\theta_w - \theta_t) - m_t L_{bar} L_{trail} \dot{\theta}_s^2 \sin(\theta_s - \theta_t) = 0_t$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_p} \right) - \frac{\partial L}{\partial x_p} = (m_p + m_j + m_t) \ddot{x}_p + (m_j + m_t) L_{hitch} (\ddot{\theta}_p \sin \theta_p + \dot{\theta}_p^2 \cos \theta_p) + (m_j + m_t) L_{bar} (\ddot{\theta}_j \sin \theta_j + \dot{\theta}_j^2 \cos \theta_j) + m_t L_{tail} (\ddot{\theta}_t \sin \theta_t + \dot{\theta}_t^2 \cos \theta_t) = Q_x$$

$$(m_j + m_t) L_{hitch} \sin \theta_p \ddot{\theta}_p + (m_j + m_t) L_{hitch} \dot{\theta}_p^2 \cos \theta_p \quad (m_j + m_t) L_{bar} \sin \theta_j \ddot{\theta}_j + (m_j + m_t) L_{bar} \dot{\theta}_j^2 \cos \theta_j \quad m_t L_{tail} \sin \theta_t \ddot{\theta}_t + m_t L_{tail} \dot{\theta}_t^2 \cos \theta_t$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}_p} \right) - \frac{\partial L}{\partial y_p} = (m_p + m_j + m_t) \ddot{y}_p - (m_j + m_t) L_{hitch} (\ddot{\theta}_p \cos \theta_p - \dot{\theta}_p^2 \sin \theta_p) - (m_j + m_t) L_{bar} (\ddot{\theta}_j \cos \theta_j - \dot{\theta}_j^2 \sin \theta_j) - m_t L_{tail} (\ddot{\theta}_t \cos \theta_t - \dot{\theta}_t^2 \sin \theta_t) = Q_y$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_p} \right) - \frac{\partial L}{\partial \theta_p} = [I_p + (m_j + m_t) L_{hitch}^2] \ddot{\theta}_p + (m_j + m_t) L_{hitch} (\ddot{x}_p \sin \theta_p - \ddot{y}_p \cos \theta_p) + (m_j + m_t) L_{hitch} L_{bar} \ddot{\theta}_j \cos(\theta_p - \theta_j) + (m_j + m_t) L_{hitch} L_{bar} \dot{\theta}_j^2 \sin(\theta_p - \theta_j) + m_t L_{hitch} L_{tail} \ddot{\theta}_t \cos(\theta_p - \theta_t) + m_t L_{hitch} L_{tail} \dot{\theta}_t^2 \sin(\theta_p - \theta_t) = Q_{\theta_p}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_j} \right) - \frac{\partial L}{\partial \theta_j} = [I_j + (m_j + m_t) L_{bar}^2] \ddot{\theta}_j + (m_j + m_t) L_{bar} (\ddot{x}_p \sin \theta_j + \ddot{y}_p \cos \theta_j) + (m_j + m_t) L_{hitch} L_{bar} \ddot{\theta}_p \cos(\theta_p - \theta_j) - (m_j + m_t) L_{hitch} L_{bar} \dot{\theta}_p^2 \sin(\theta_p - \theta_j) + m_t L_{bar} L_{tail} \ddot{\theta}_t \cos(\theta_j - \theta_t) + m_t L_{bar} L_{tail} \dot{\theta}_t^2 \sin(\theta_j - \theta_t) = Q_{\theta_j}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_t} \right) - \frac{\partial L}{\partial \theta_t} = [I_t + m_t L_{tail}^2] \ddot{\theta}_t + m_t L_{tail} (\ddot{x}_p \sin \theta_t - \ddot{y}_p \cos \theta_t) + m_t L_{hitch} L_{tail} \ddot{\theta}_p \cos(\theta_p - \theta_t) + m_t L_{bar} L_{tail} \ddot{\theta}_j \cos(\theta_j - \theta_t) - m_t L_{hitch} L_{tail} \dot{\theta}_p^2 \sin(\theta_p - \theta_t) - m_t L_{bar} L_{tail} \dot{\theta}_j^2 \sin(\theta_j - \theta_t) = Q_{\theta_t}$$

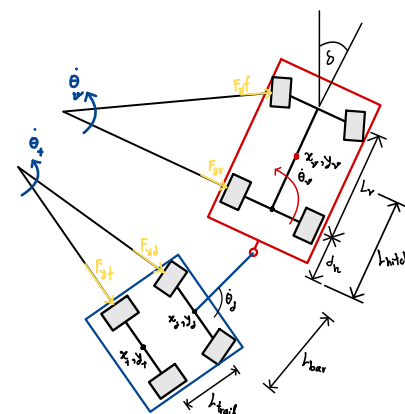
Multi-body Rigid Body Dynamic Form

$$M(q) \ddot{q} + n(q, \dot{q}) = Q$$

$$M(q) \ddot{q} = \begin{bmatrix} M_{11} & M_{12} & M_{13} & M_{14} & M_{15} \\ M_{21} & M_{22} & M_{23} & M_{24} & M_{25} \\ M_{31} & M_{32} & M_{33} & M_{34} & M_{35} \\ M_{41} & M_{42} & M_{43} & M_{44} & M_{45} \\ M_{51} & M_{52} & M_{53} & M_{54} & M_{55} \end{bmatrix} \begin{bmatrix} \ddot{x}_p \\ \ddot{y}_p \\ \ddot{\theta}_p \\ \ddot{\theta}_j \\ \ddot{\theta}_t \end{bmatrix} \begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \end{bmatrix}$$

$$n(q, \dot{q}) = \begin{bmatrix} n_x \\ n_y \\ n_{\theta_p} \\ n_{\theta_j} \\ n_{\theta_t} \end{bmatrix} = \begin{bmatrix} (m_j + m_t) L_{hitch} \dot{\theta}_p^2 \cos \theta_p + (m_j + m_t) L_{bar} \dot{\theta}_j^2 \cos \theta_j + m_t L_{tail} \dot{\theta}_t^2 \cos \theta_t \\ (m_j + m_t) L_{hitch} \dot{\theta}_p^2 \sin \theta_p + (m_j + m_t) L_{bar} \dot{\theta}_j^2 \sin \theta_j + m_t L_{tail} \dot{\theta}_t^2 \sin \theta_t \\ (m_j + m_t) L_{hitch} L_{bar} \dot{\theta}_p^2 \sin(\theta_p - \theta_j) + m_t L_{hitch} L_{tail} \dot{\theta}_p^2 \sin(\theta_p - \theta_t) \\ (m_j + m_t) L_{hitch} L_{bar} \dot{\theta}_p^2 \sin(\theta_p - \theta_j) + m_t L_{bar} L_{tail} \dot{\theta}_j^2 \sin(\theta_j - \theta_t) \\ -m_t L_{hitch} L_{tail} \dot{\theta}_p^2 \sin(\theta_p - \theta_t) - m_t L_{bar} L_{tail} \dot{\theta}_j^2 \sin(\theta_j - \theta_t) \end{bmatrix}$$

$$\begin{bmatrix} (m_p + m_j + m_t) & 0 & (m_j + m_t) L_{hitch} \sin \theta_p & (m_j + m_t) L_{bar} \sin \theta_j & m_t L_{tail} \sin \theta_t \\ 0 & m_p + m_j + m_t & (m_j + m_t) L_{hitch} \cos \theta_p & -(m_j + m_t) L_{bar} \cos \theta_j & -m_t L_{tail} \cos \theta_t \\ (m_j + m_t) L_{hitch} \sin \theta_p & (m_j + m_t) L_{hitch} \cos \theta_p & I_p + (m_j + m_t) L_{hitch}^2 & (m_j + m_t) L_{hitch} L_{bar} \cos(\theta_p - \theta_j) & m_t L_{hitch} L_{tail} \cos(\theta_p - \theta_t) \\ (m_j + m_t) L_{bar} \sin \theta_j & -(m_j + m_t) L_{bar} \cos \theta_j & (m_j + m_t) L_{hitch} L_{bar} \cos(\theta_p - \theta_j) & I_j + (m_j + m_t) L_{bar}^2 & m_t L_{bar} L_{tail} \cos(\theta_j - \theta_t) \\ m_t L_{tail} \sin \theta_t & -m_t L_{tail} \cos \theta_t & m_t L_{hitch} L_{tail} \cos(\theta_p - \theta_t) & m_t L_{bar} L_{tail} \cos(\theta_j - \theta_t) & I_t + m_t L_{tail}^2 \end{bmatrix} \begin{bmatrix} \ddot{x}_p \\ \ddot{y}_p \\ \ddot{\theta}_p \\ \ddot{\theta}_j \\ \ddot{\theta}_t \end{bmatrix} + \begin{bmatrix} (m_j + m_t) L_{hitch} \dot{\theta}_p^2 \cos \theta_p + (m_j + m_t) L_{bar} \dot{\theta}_j^2 \cos \theta_j + m_t L_{tail} \dot{\theta}_t^2 \cos \theta_t \\ (m_j + m_t) L_{hitch} \dot{\theta}_p^2 \sin \theta_p + (m_j + m_t) L_{bar} \dot{\theta}_j^2 \sin \theta_j + m_t L_{tail} \dot{\theta}_t^2 \sin \theta_t \\ (m_j + m_t) L_{hitch} L_{bar} \dot{\theta}_p^2 \sin(\theta_p - \theta_j) + m_t L_{hitch} L_{tail} \dot{\theta}_p^2 \sin(\theta_p - \theta_t) \\ (m_j + m_t) L_{hitch} L_{bar} \dot{\theta}_p^2 \sin(\theta_p - \theta_j) + m_t L_{bar} L_{tail} \dot{\theta}_j^2 \sin(\theta_j - \theta_t) \\ -m_t L_{hitch} L_{tail} \dot{\theta}_p^2 \sin(\theta_p - \theta_t) - m_t L_{bar} L_{tail} \dot{\theta}_j^2 \sin(\theta_j - \theta_t) \end{bmatrix} = \begin{bmatrix} Q_x \\ Q_y \\ Q_{\theta_p} \\ Q_{\theta_j} \\ Q_{\theta_t} \end{bmatrix}$$



$$\begin{bmatrix} (m_p + m_j + m_t) & 0 & (m_j + m_t) L_{hitch} \sin \theta_p & (m_j + m_t) L_{bar} \sin \theta_j & m_t L_{tail} \sin \theta_t \\ 0 & m_p + m_j + m_t & (m_j + m_t) L_{hitch} \cos \theta_p & -(m_j + m_t) L_{bar} \cos \theta_j & -m_t L_{tail} \cos \theta_t \\ (m_j + m_t) L_{hitch} \sin \theta_p & (m_j + m_t) L_{hitch} \cos \theta_p & I_p + (m_j + m_t) L_{hitch}^2 & (m_j + m_t) L_{hitch} L_{bar} \cos(\theta_p - \theta_j) & m_t L_{hitch} L_{tail} \cos(\theta_p - \theta_t) \\ (m_j + m_t) L_{bar} \sin \theta_j & -(m_j + m_t) L_{bar} \cos \theta_j & (m_j + m_t) L_{hitch} L_{bar} \cos(\theta_p - \theta_j) & I_j + (m_j + m_t) L_{bar}^2 & m_t L_{bar} L_{tail} \cos(\theta_j - \theta_t) \\ m_t L_{tail} \sin \theta_t & -m_t L_{tail} \cos \theta_t & m_t L_{hitch} L_{tail} \cos(\theta_p - \theta_t) & m_t L_{bar} L_{tail} \cos(\theta_j - \theta_t) & I_t + m_t L_{tail}^2 \end{bmatrix} \begin{bmatrix} \ddot{x}_p \\ \ddot{y}_p \\ \ddot{\theta}_p \\ \ddot{\theta}_j \\ \ddot{\theta}_t \end{bmatrix} + \begin{bmatrix} (m_j + m_t) L_{hitch} \dot{\theta}_p^2 \cos \theta_p + (m_j + m_t) L_{bar} \dot{\theta}_j^2 \cos \theta_j + m_t L_{tail} \dot{\theta}_t^2 \cos \theta_t \\ (m_j + m_t) L_{hitch} \dot{\theta}_p^2 \sin \theta_p + (m_j + m_t) L_{bar} \dot{\theta}_j^2 \sin \theta_j + m_t L_{tail} \dot{\theta}_t^2 \sin \theta_t \\ (m_j + m_t) L_{hitch} L_{bar} \dot{\theta}_p \dot{\theta}_j \sin(\theta_p - \theta_j) + m_t L_{hitch} L_{tail} \dot{\theta}_p \dot{\theta}_t \sin(\theta_p - \theta_t) \\ (m_j + m_t) L_{hitch} L_{bar} \dot{\theta}_p \dot{\theta}_j \sin(\theta_p - \theta_j) + m_t L_{bar} L_{tail} \dot{\theta}_j \dot{\theta}_t \sin(\theta_j - \theta_t) \\ -m_t L_{hitch} L_{tail} \dot{\theta}_p \dot{\theta}_t \sin(\theta_p - \theta_t) - m_t L_{bar} L_{tail} \dot{\theta}_j \dot{\theta}_t \sin(\theta_j - \theta_t) \end{bmatrix} = \begin{bmatrix} F_x - F_j \sin \delta - R_{xf} - R_{xr} - R_a - R_j \cos(\theta_p - \theta_j) - W \sin \theta_p \\ F_y \cos \delta + F_{yr} - R_j \sin(\theta_p - \theta_j) \\ F_j \cos \delta - R_{jf} - F_{jr} + R_j \sin(\theta_p - \theta_j) L_{hitch} \\ -F_j L_{bar} - F_{jt} L_{bar} \cos(\theta_j - \theta_t) - (R_{jt} + Y_t \sin \theta_j) \sin(\theta_j - \theta_t) \\ -F_t L_{tail} \end{bmatrix}$$